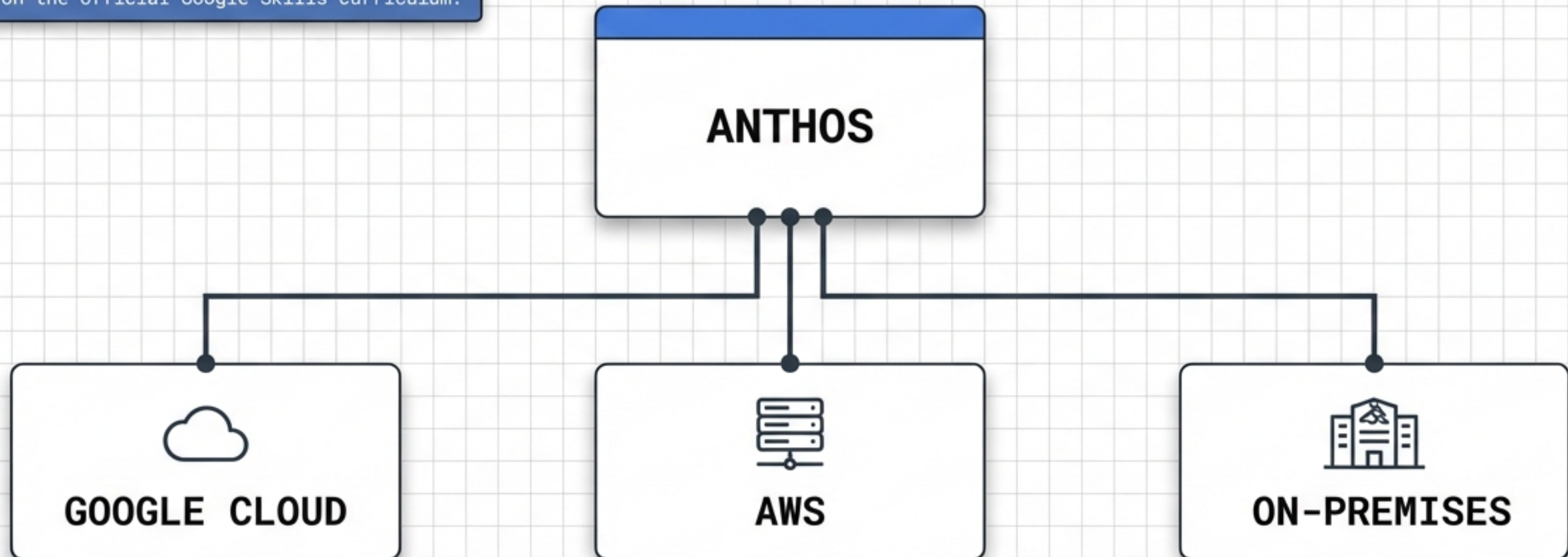


Hybrid Cloud Infrastructure Foundations with Anthos

Master modern application management across on-premises and multi-cloud environments.

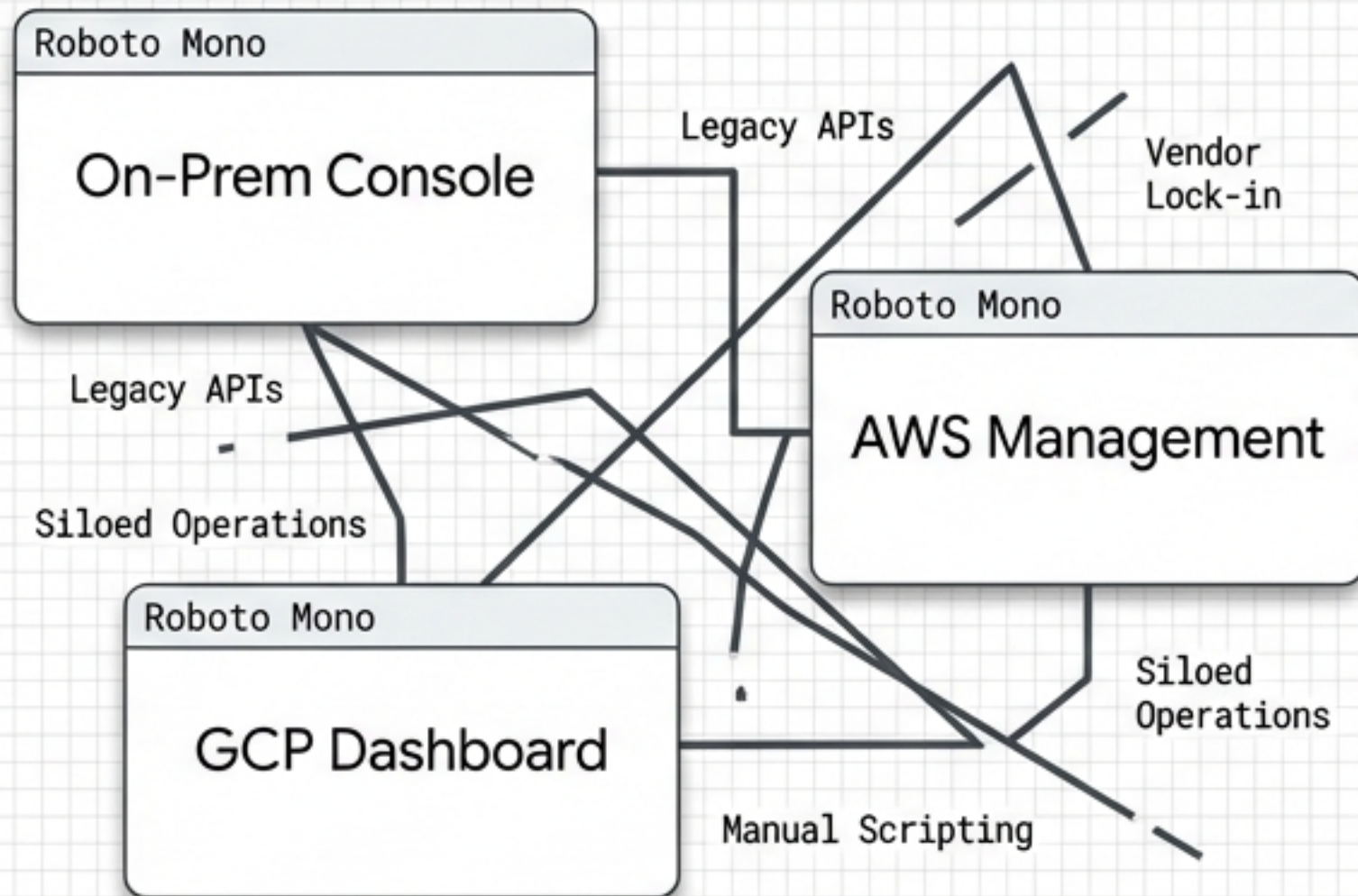
Based on the official Google Skills curriculum.



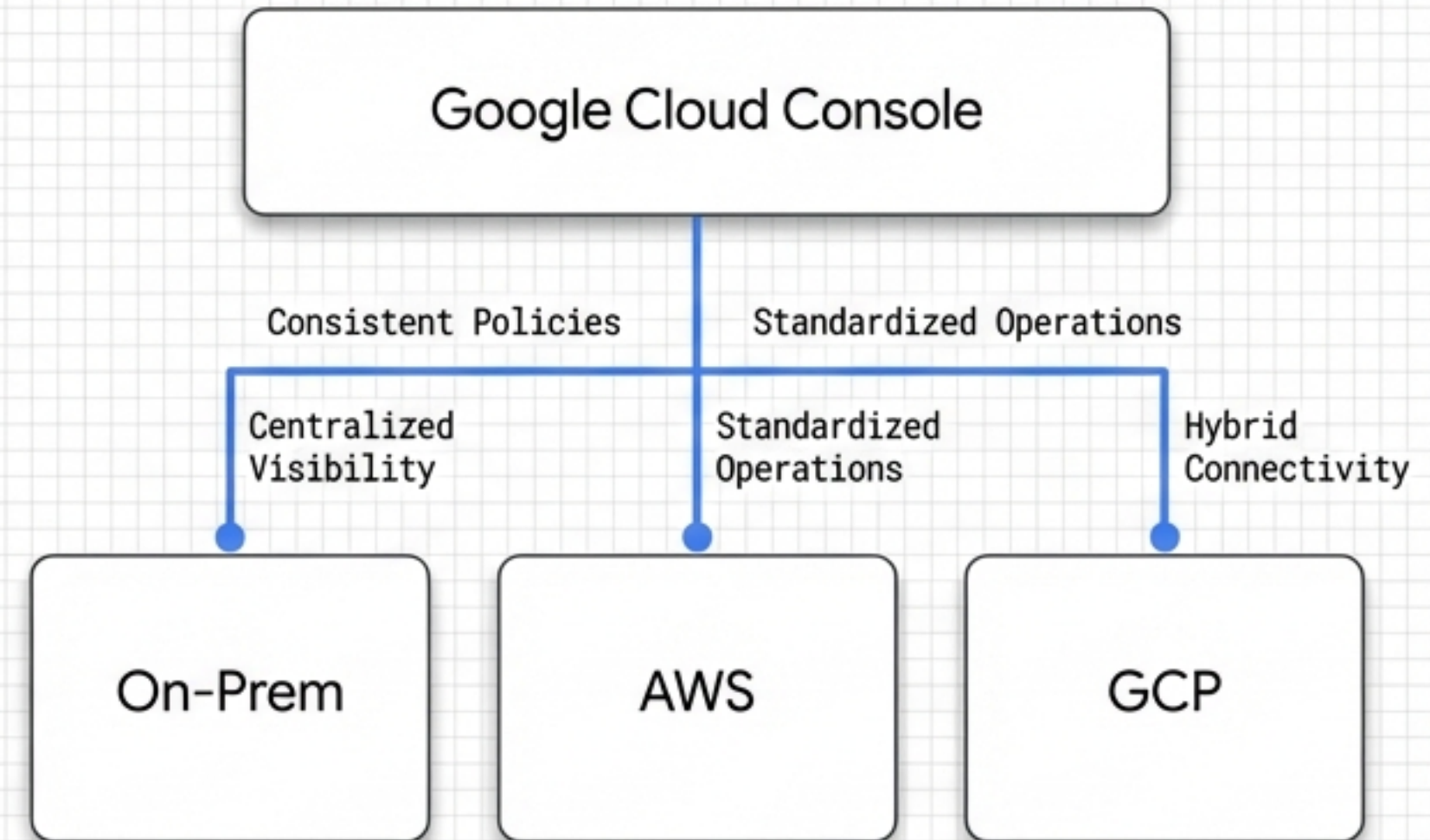
The Promise of One Consistent Experience

Anthos gives architects and administrators the freedom to build, deploy, and manage modern applications anywhere—eliminating the friction of disjointed platforms and vendor-specific silos.

Fragmented Reality

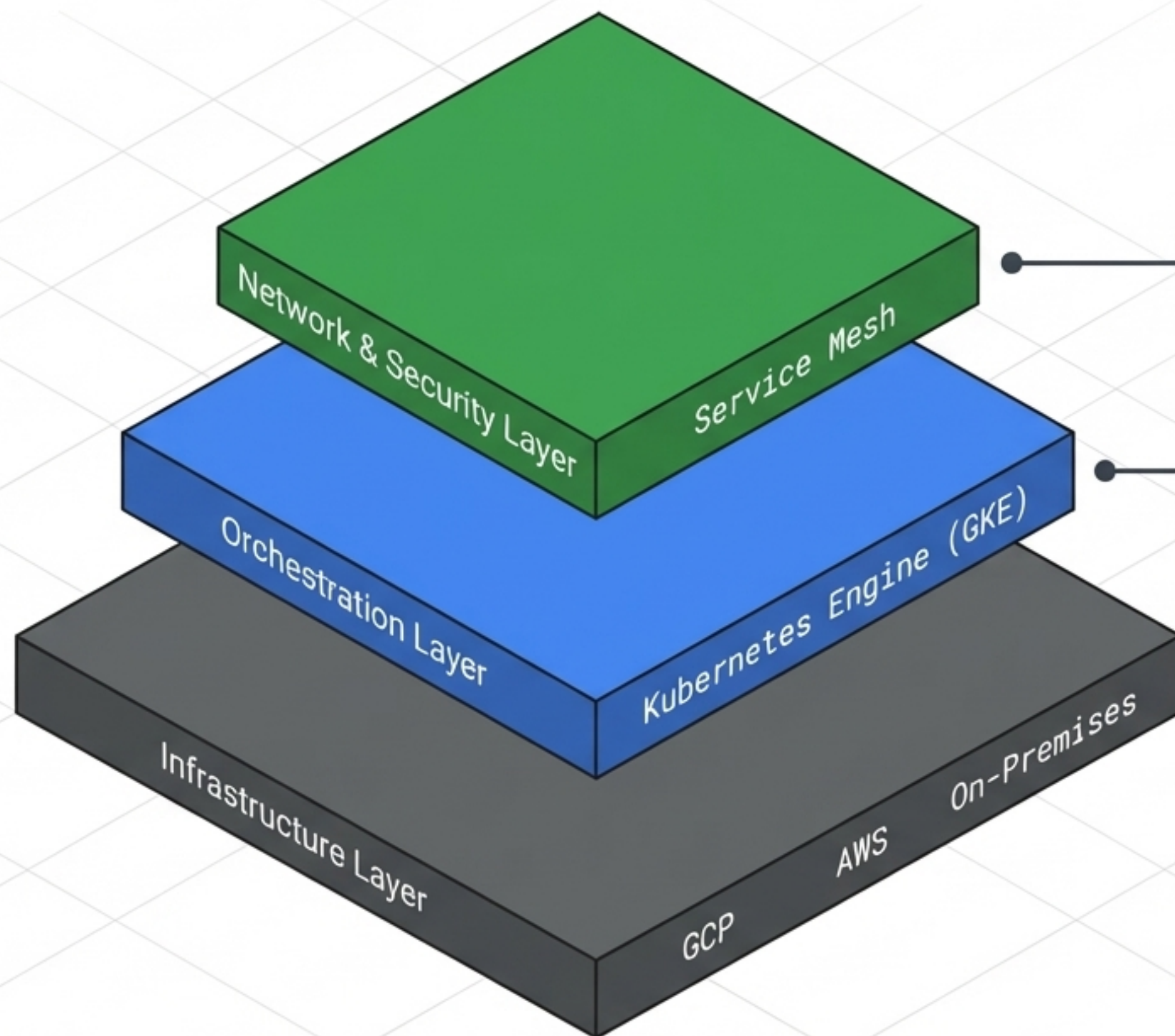


Unified Control Plane



Move from environment-specific chaos to a unified, hybrid control plane.

Deconstructing the Anthos Technology Stack



Module 2: Managing Hybrid Clusters.

Learn to deploy and scale GKE On-Prem architecture across diverse physical and cloud environments.

Module 3: Introduction to Service Mesh.

Master the network fabric that connects, secures, and monitors your microservices regardless of where they run.

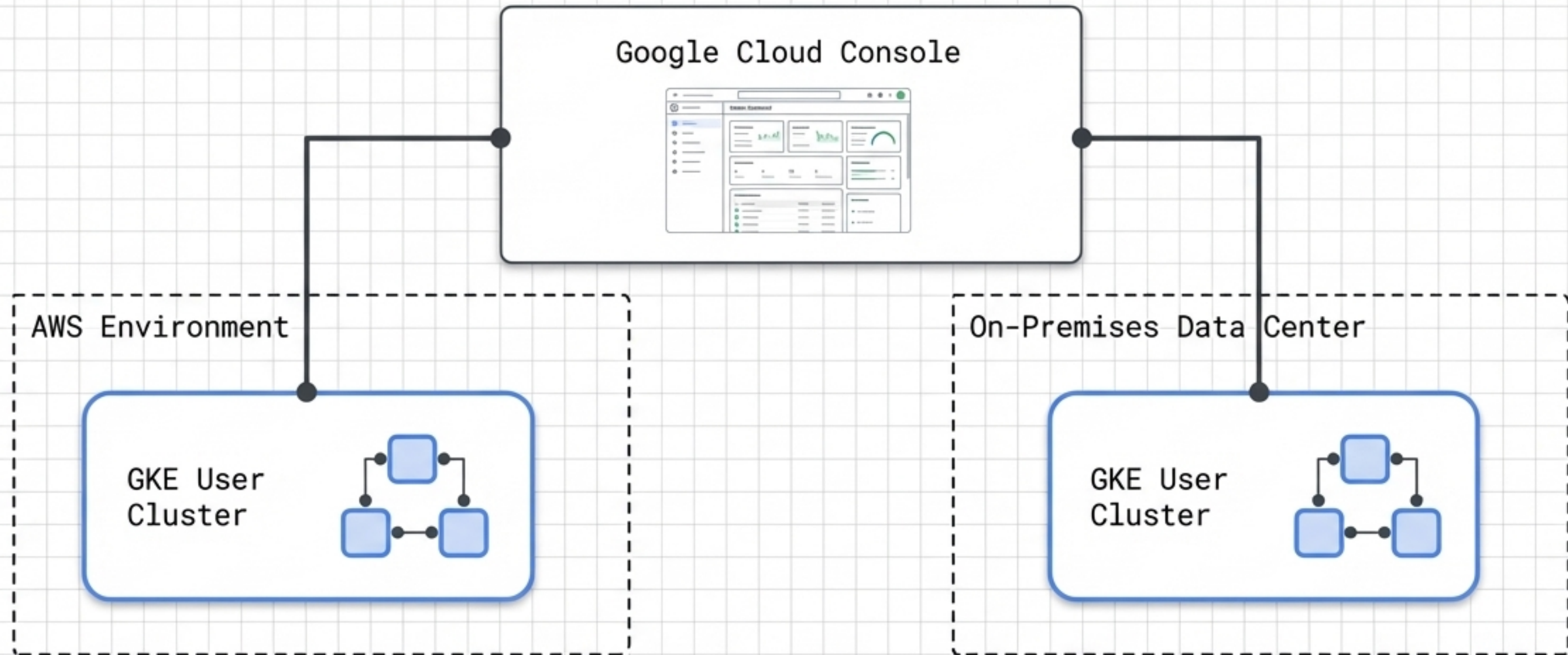
The Shift to Modern Solutions

Dimension	Fragmented Infrastructure	Anthos Hybrid Cloud
Operations	Multiple, disjointed proprietary tools.	Centralized Google Cloud Console.
Portability	Heavy vendor lock-in and refactoring.	“Write once, run anywhere” containerization.
Networking	Manual routing and firewall rules.	Automated, policy-driven Service Mesh.
Environment	Siloed On-Premises & Public Clouds.	Unified, location-agnostic infrastructure.

Containers and Kubernetes Everywhere

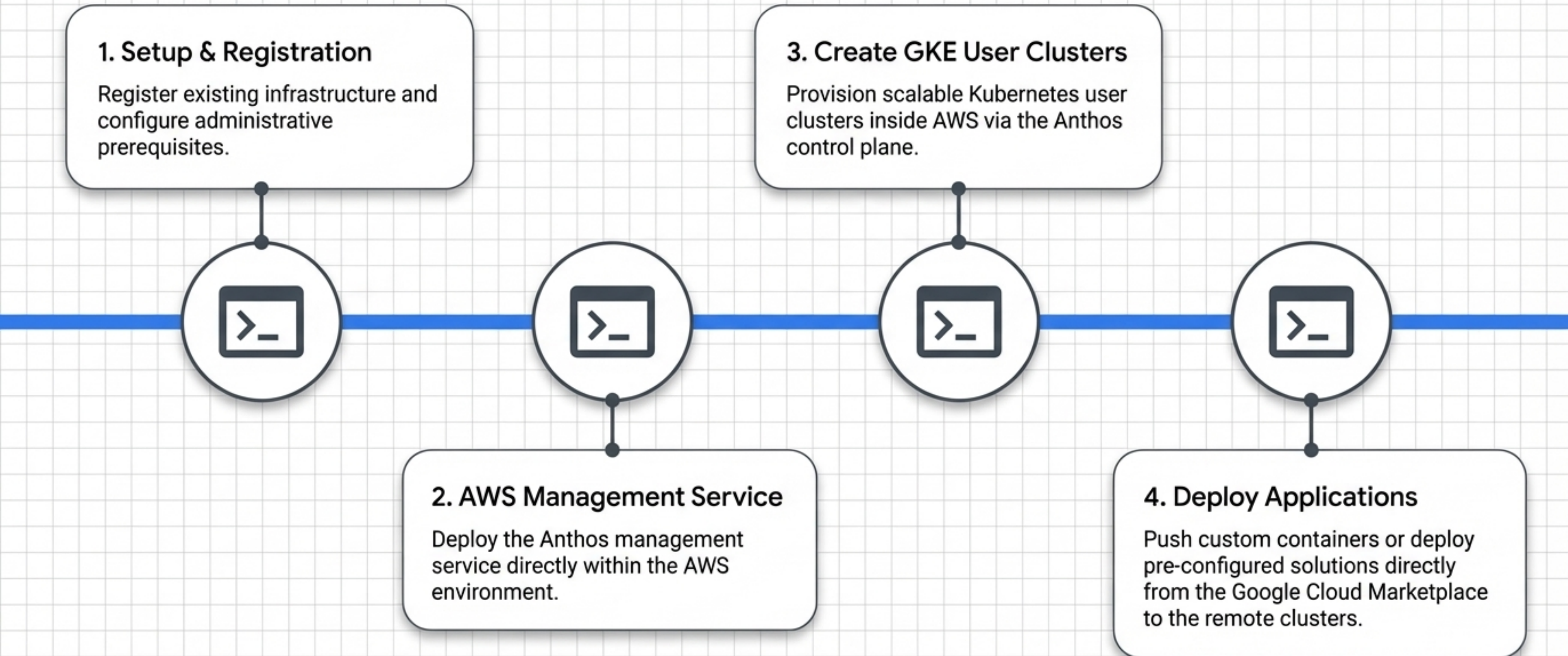
Core Concept: Anthos extends the power of Google Kubernetes Engine (GKE) beyond Google Cloud, deploying managed user clusters directly into your existing AWS or On-Premises infrastructure.

Key Focus: GKE On-Prem Architecture & Hybrid Networking basics.



Administrators manage all remote clusters seamlessly through a single, centralized Google Cloud interface, regardless of physical location.

From Setup to Production: The Anthos Lab Workflow



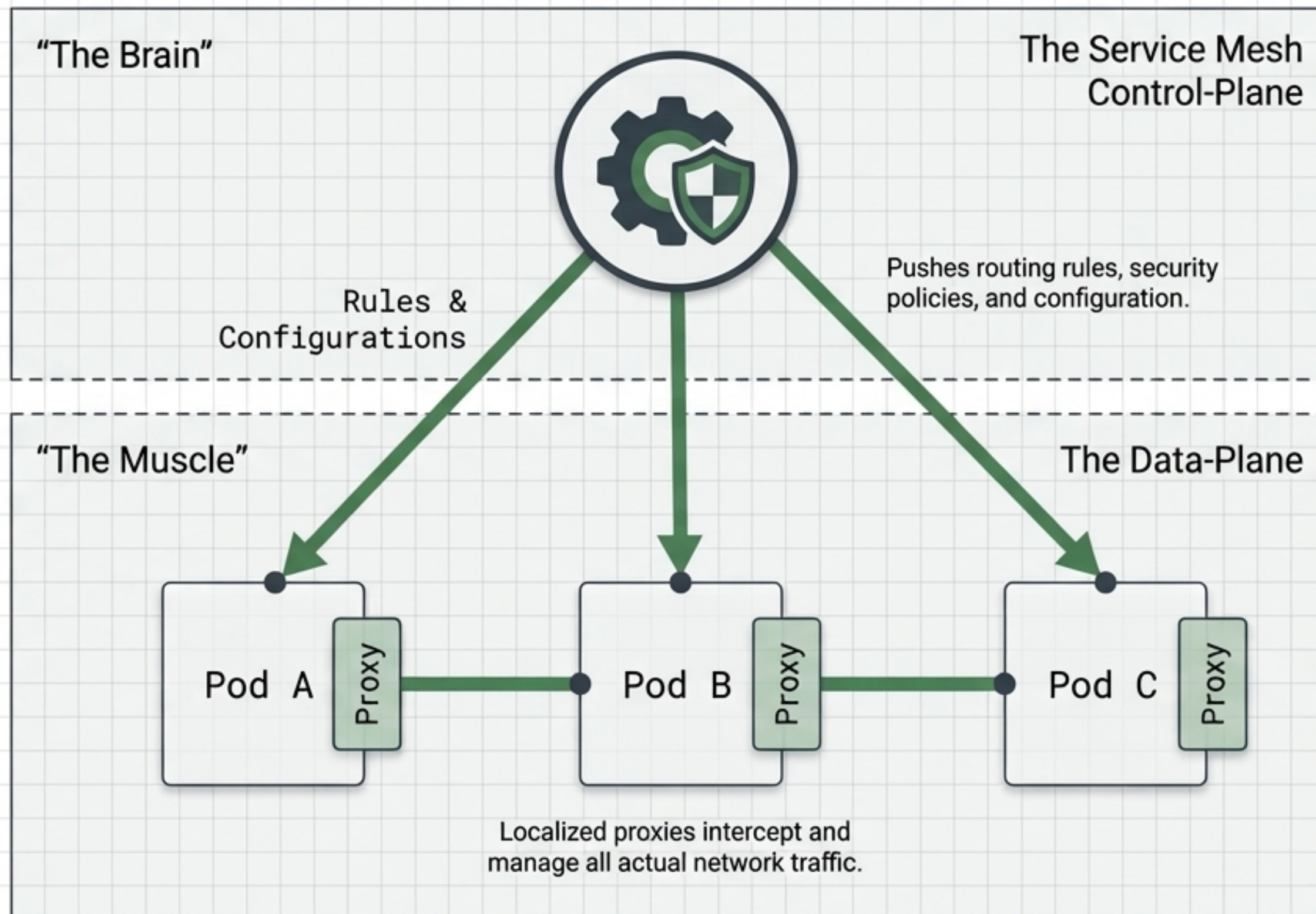
Demystifying the Service Mesh

Why a Service Mesh?

An essential network fabric that ensures microservices can communicate reliably without embedding network logic into the application code.

Core Benefits:

Unlocks system-wide Observability, Security, and Reliability.



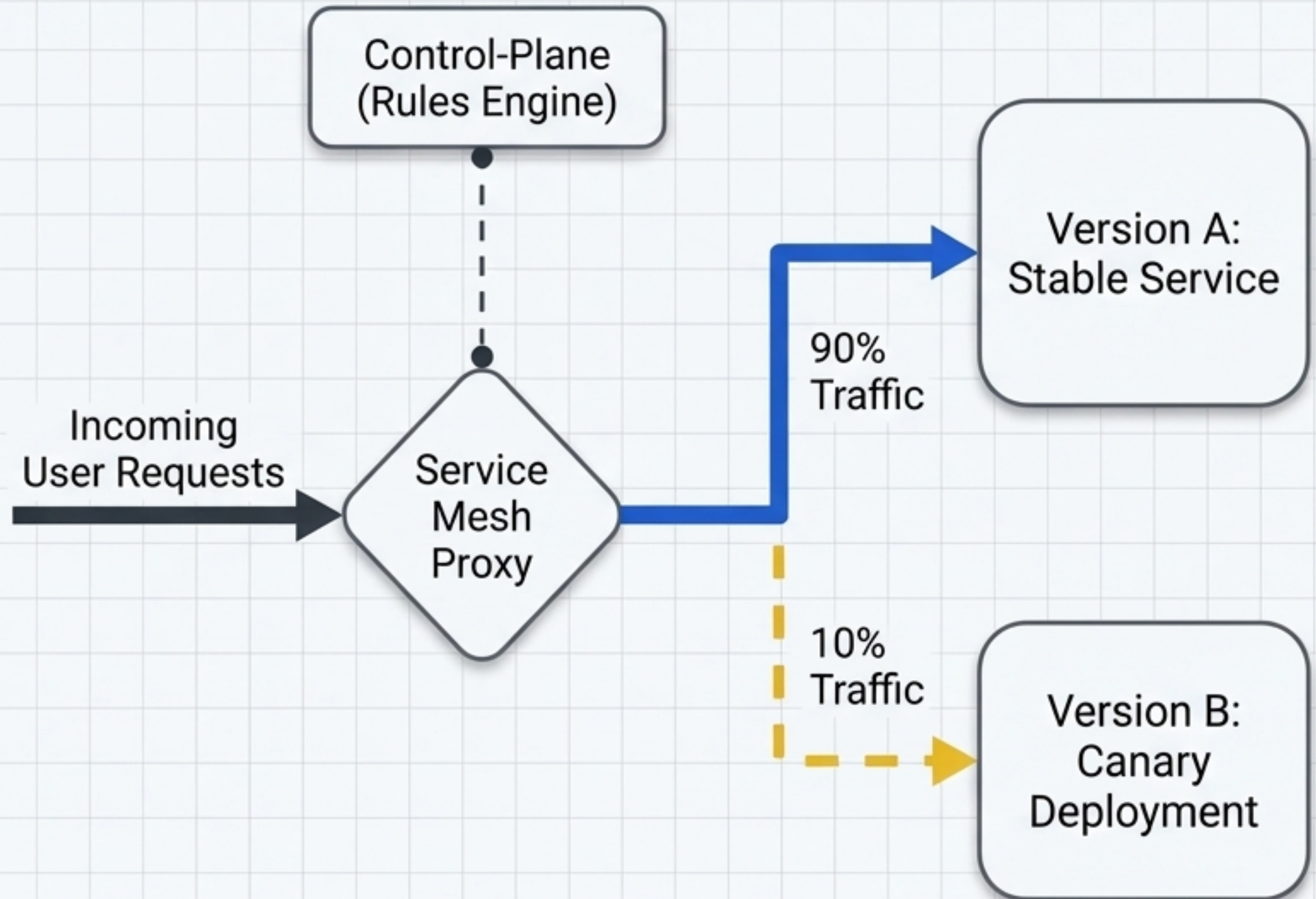
Intelligent Routing at the Network Edge

Request Flow:

Every inbound request hits the proxy first, allowing the Service Mesh to intelligently decide its destination based on real-time rules, not static IP addresses.

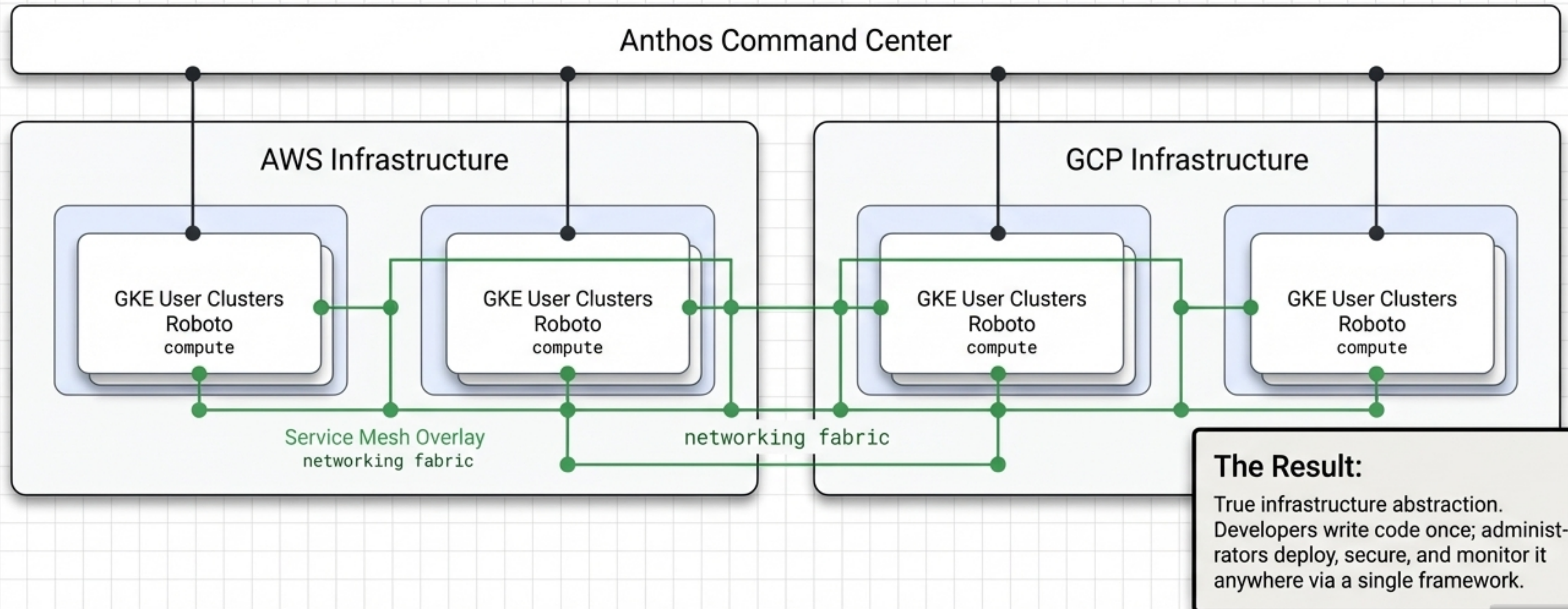
Intro to Traffic Shaping:

Administrators can split traffic dynamically. Route 10% of users to a new canary release, inject artificial faults for testing, or automatically reroute around failing nodes—all without touching application code.



The Complete Hybrid Architecture

Synthesis Insight: Anthos is not a single tool—it is the seamless orchestration of **Kubernetes** (compute capabilities) and **Service Mesh** (networking fabric) across any underlying infrastructure.



Validate Your Architectural Skills

